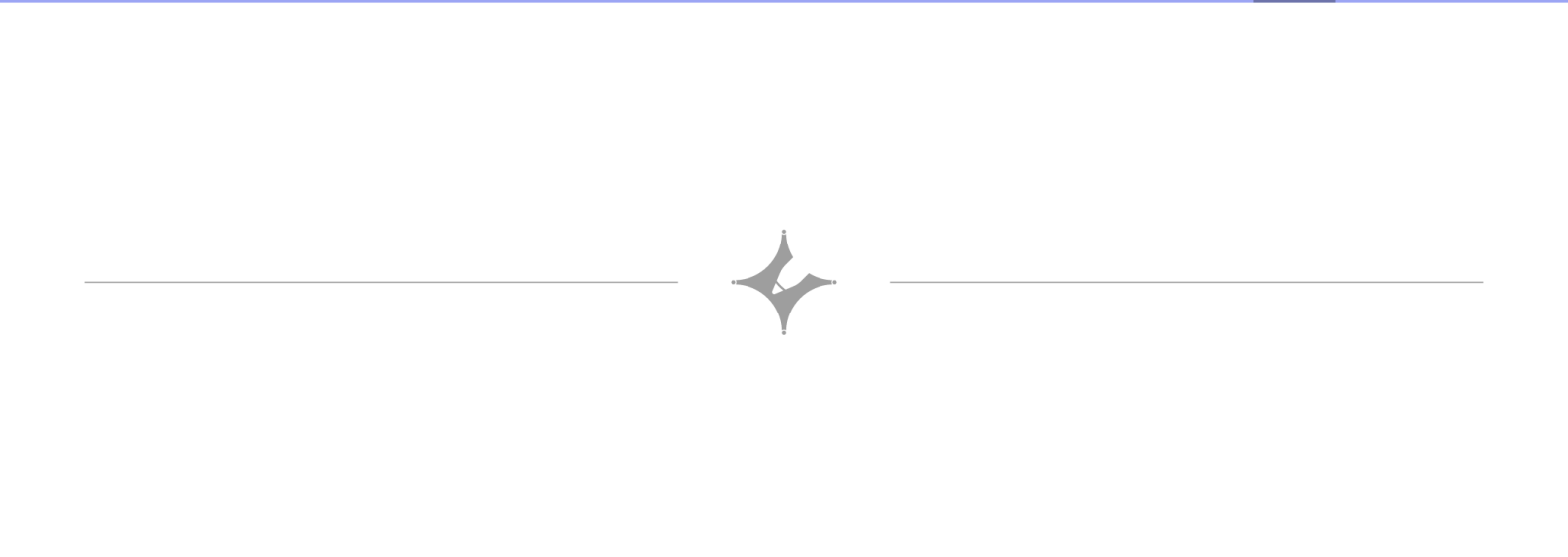
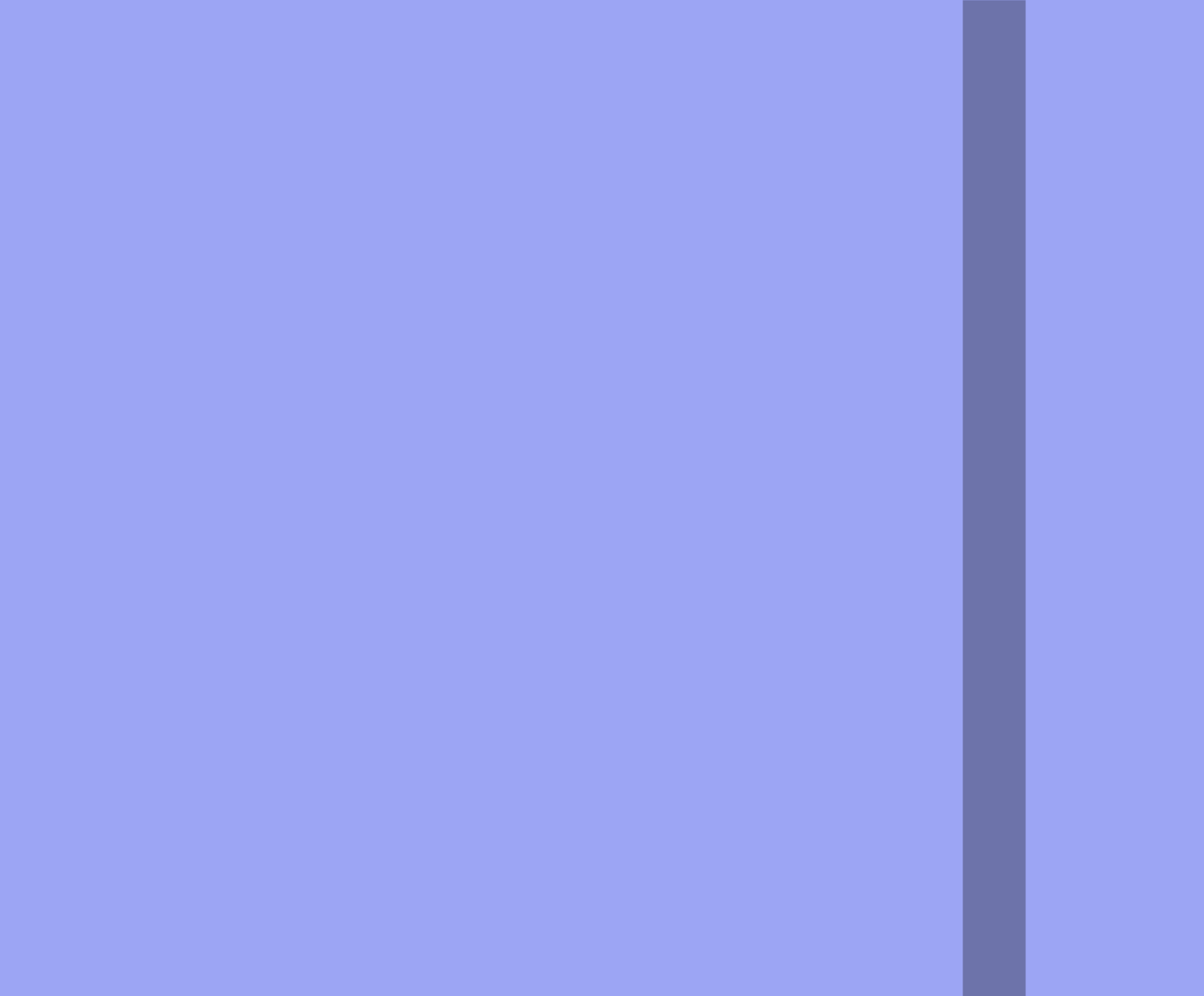




a TM M that print $\langle M \rangle$ on the tape

$$M: A \rightarrow B.$$

A writes $\langle B \rangle$ to the tape

$\langle B \rangle$

B writes $\langle A \rangle$ to the tape
and swap it with $\langle B \rangle$.

function $q: \{0,1\}^* \rightarrow \{0,1\}^*$, $q(w) = \langle M_w \rangle$

✓
 q is computable

$M_w =$ on input x

1. write w to the tape

$B =$ on input w

1. compute $q(w)$

2. write $q(w)$ to the tape

and swap($q(w)$, w).

$A =$ on any input

1. erase the input

2. write $\langle B \rangle$ to the tape

run A: $\boxed{\langle B \rangle}$ $\rightarrow$ run B: $\boxed{q(\langle B \rangle) \langle B \rangle}$

$q(\langle B \rangle)$ 就是 $\langle A \rangle$!!

定理 1: Recursion Theorem

For any TM T , that takes an input $\langle M \rangle$ and x ,

$\exists$ TM R , s.t. $R(x) = T(\langle R \rangle, x)$.

证:

T

$R: A \rightarrow B \rightarrow T$

A writes $\langle B \rangle \langle T \rangle$ to the tape

$\boxed{x \langle B \rangle \langle T \rangle}$

B writes $\langle A \rangle$ to the tape
and reorder.

$\boxed{\underbrace{\langle A \rangle \langle B \rangle \langle T \rangle}_R x}$

$T(\langle R \rangle, x)$.

#.

例:

可以用 Rec Theorem 证明 halt:

反证: 设 HALT 可计算:

$M =$ on input w

1. obtain $\langle M \rangle$.
2. compute $\text{HALT}(M, w)$
3. if HALT:
 loop
 else:
 halt

矛盾.

#.

例:

We say TM M 最小 if $\forall M' \Leftrightarrow M, |\langle M' \rangle| \geq |\langle M \rangle|$.

$\text{MINI}(M)$ 不可计算.

证: 反证.

$M =$ on input x

1. obtain $\langle M \rangle$.
2. for s in $\{0, 1\}^*$, 升序.

if $MIN(S) = 1$ and $|S| > |\langle M \rangle|$.
 run S on x
 return $S(x)$

$M \Leftrightarrow S$ 但 $|\langle M \rangle| < |S|$

$\Downarrow$
 S 非最小
 矛盾.

#.

例: (fixed pointed) 定理.

let t 可计算, $\exists TM.F$, s.t. $t(\langle F \rangle) = \langle M \rangle$,
 且 $M \Leftrightarrow F$

证:

$F =$ on input w

1. obtain $\langle F \rangle$.
2. compute $t(\langle F \rangle)$, let $\langle M \rangle = t(\langle F \rangle)$
3. simulate M on w .

哥德尔不完备定理.

let $T \subseteq \{0,1\}^*$ be a set (set of true statement.

A proof system for T is a Turing Machine V that satisfies:

(1). (effectiveness). $\forall x, t \in \{0,1\}^*$
 $V(x, t)$ halt, return 0/1

(2). (soundness) $\forall x \notin T, \forall t \in \{0,1\}^*$
 $V(x, t) = 0$

We say V 完备 if
(complete)
 $\forall x \in T, \exists t, V(x, t) = 1$

定理:

$\exists T$, 无完备证明系统

证:

T = set of all true statements of:

"a TM M halts on 0" or

"a TM M not halts on 0 "

if T 有完备系统 V

并行做:

- run M on 0
- let $x = "M \text{ does not halt on } 0"$
- for $t \in \{0, 1\}^*$ 升序
- run $V(x, t)$.
- if $V(x, t) == 1$
- return 0

必有一个停机

#.

Quantified Integer Statement

符号

$+$	x	$=$
$\wedge$	$\vee$	$\rightarrow$
$\forall$	$\exists$	(quantifier)

Given a QIS Φ

$$\text{EVAL}(\phi) = \begin{cases} 1 & \text{if } \phi \text{ is true} \\ 0 & \text{false.} \end{cases}$$

不可计算.

定理: let T be the set of QIS,

T 无完备 proof system.